

CAP-482 DATA EXPLORATION

CA-01

DATE- 26-02-2026

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Registration no. - 12400111

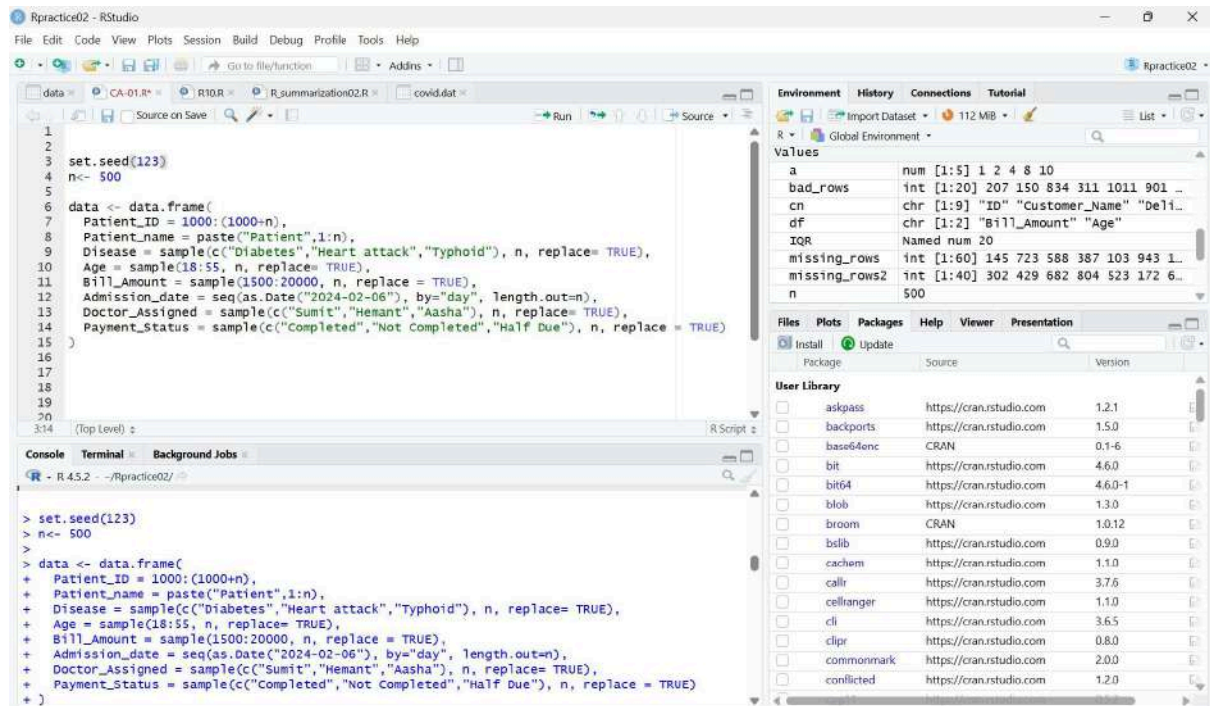
Roll no.-47, group02

Section- D2412

Create Synthetic dataset(using dataframe)

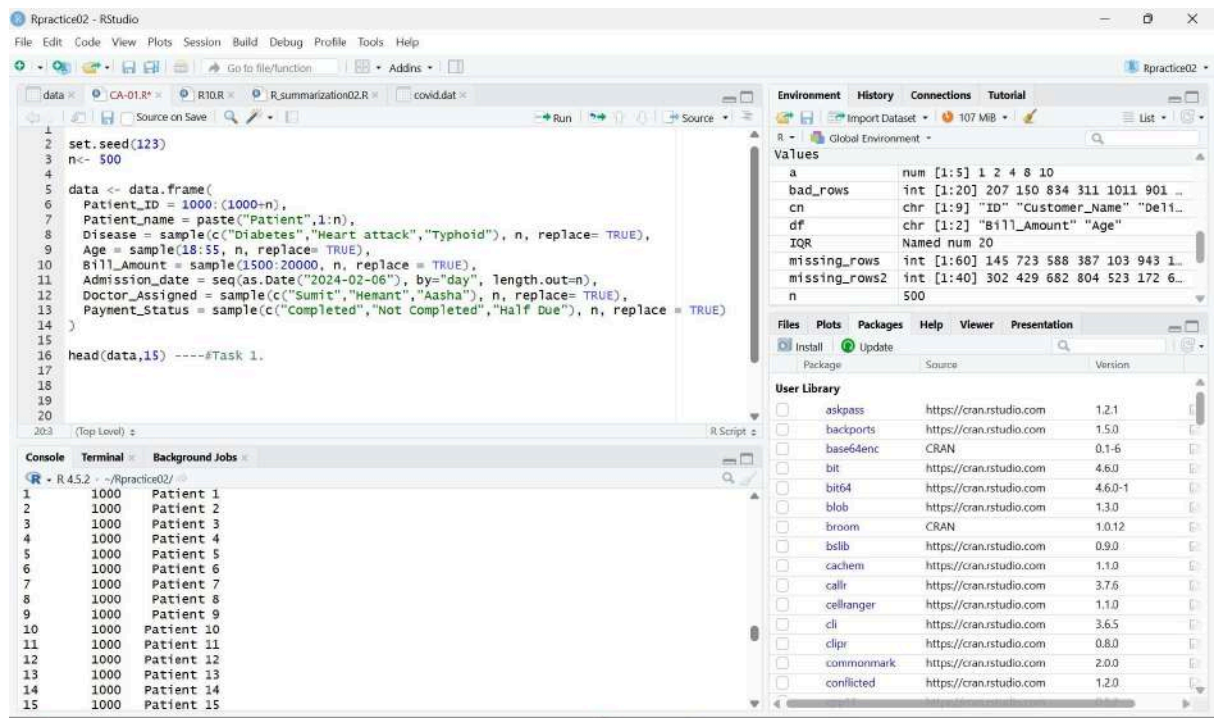
Create a synthetic dataset containing at least 500 records with the following fields:

- Patient_ID
- Patient_Name
- Disease
- Age
- Bill_Amount
- Admission_Date
- Doctor_Assigned
- Payment_Status



Data understanding

1. Display the first 15 rows



2. Check number of rows and columns

The screenshot shows the RStudio interface with the following R code in the script editor:

```
data <- data.frame(
  Patient_ID = 1000:(1000+n),
  Patient_name = paste("Patient",1:n),
  Disease = sample(c("Diabetes","Heart attack","Typhoid"), n, replace= TRUE),
  Age = sample(18:55, n, replace= TRUE),
  Bill_Amount = sample(1500:20000, n, replace= TRUE),
  Admission_date = seq(as.Date("2024-02-06"), by="day", length.out=n),
  Doctor_Assigned = sample(c("Sumit","Hemant","Aasha"), n, replace= TRUE),
  Payment_Status = sample(c("Completed","Not Completed","Half Due"), n, replace= TRUE)
)

head(data,15) ----#Task 1.

dim(data) -----#Task 2.

View(data)

arrange(data,desc("Bill_Amount"))
```

The console output shows the first 15 rows of the data frame:

```
5 Completed
6 Not Completed
7 Half Due
8 Completed
9 Half Due
10 Half Due
11 Half Due
12 Completed
13 Completed
14 Half Due
15 Not Completed
> dim(data)
[1] 500 8
>
```

The Environment pane on the right shows the variables in the Global Environment:

Variable	Class	Length	Values
a	num	[1:5]	1 2 4 8 10
bad_rows	int	[1:20]	207 150 834 311 1011 901 ...
cn	chr	[1:9]	"ID" "Customer_Name" "Deliv...
df	chr	[1:2]	"Bill_Amount" "Age"
IQR	Named num	20	
missing_rows	int	[1:60]	145 723 588 387 103 943 1...
missing_rows2	int	[1:40]	302 429 682 804 523 172 6...
n	int		500

3. Display column data types

The screenshot shows the RStudio interface with the same R code as in the previous screenshot. The console output now shows the data types for the first few rows of the data frame:

```
'data.frame': 500 obs. of 8 variables:
 $ Patient_ID : int 1000 1000 1000 1000 1000 1000 1000 1000 1000 ...
 $ Patient_name : chr "Patient 1" "Patient 2" "Patient 3" "Patient 4" ...
 $ Disease : chr "Typhoid" "Typhoid" "Typhoid" "Heart attack" ...
 $ Age : int 37 19 27 24 55 53 40 51 41 21 ...
 $ Bill_Amount : int 9288 17737 13946 3201 10519 14618 5797 16772 16265 18647 ...
 $ Admission_date : Date, format: "2024-02-06" "2024-02-07" "2024-02-08" ...
 $ Doctor_Assigned: chr "Hemant" "Sumit" "Aasha" "Sumit" ...
 $ Payment_Status : chr "Not Completed" "Completed" "Not Completed" "Half Due" ...

'data.frame': 500 obs. of 8 variables:
 $ Patient_ID : int 1000 1000 1000 1000 1000 1000 1000 1000 1000 ...
 $ Patient_name : chr "Patient 1" "Patient 2" "Patient 3" "Patient 4" ...
 $ Disease : chr "Typhoid" "Typhoid" "Typhoid" "Heart attack" ...
 $ Age : int 37 19 27 24 55 53 40 51 41 21 ...
 $ Bill_Amount : int 9288 17737 13946 3201 10519 14618 5797 16772 16265 18647 ...
```

The Environment pane on the right shows the variables in the Global Environment with their data types and lengths:

Variable	Class	Length	Values
a	num	[1:5]	1 2 4 8 10
bad_rows	int	[1:20]	207 150 834 311 1011 901 ...
cn	chr	[1:9]	"ID" "Customer_Name" "Deliv...
df	chr	[1:2]	"Bill_Amount" "Age"
IQR	Named num	20	
missing_rows	int	[1:60]	145 723 588 387 103 943 1...
missing_rows2	int	[1:40]	302 429 682 804 523 172 6...
n	int		500

4. Generate statistical summary

The screenshot shows the RStudio interface. The script editor contains the following code:

```
data <- data.frame(
  Patient_ID = 1000:(1000+n),
  Patient_name = paste("Patient",1:n),
  Disease = sample(c("Diabetes","Heart attack","Typhoid"), n, replace= TRUE),
  Age = sample(18:55, n, replace= TRUE),
  Bill_Amount = sample(1500:20000, n, replace= TRUE),
  Admission_date = seq(as.Date("2024-02-06"), by="day", length.out=n),
  Doctor_Assigned = sample(c("Sumit","Hemant","Aasha"), n, replace= TRUE),
  Payment_Status = sample(c("Completed","Not Completed","Half Due"), n, replace= TRUE)
)
head(data,15) ----#Task 1.
dim(data) -----#Task 2.
str(data) -----#task 3.
summary(data)
```

The console output shows the summary of the data frame:

```
> summary(data)
  Patient_ID Patient_name      Disease      Age      Bill_Amount
Min.   :1000      Length:500      Min.   :18.00      Min.   : 1548
1st Qu.:1000      Class :character 1st Qu.:27.00      1st Qu.: 6348
Median :1000      Mode  :character  Median :37.00      Median :10618
Mean   :1000      Mean   :36.55      Mean   :10693
3rd Qu.:1000      3rd Qu.:46.00      3rd Qu.:15074
Max.   :1000      Max.   :55.00      Max.   :19991

Admission_date Doctor_Assigned Payment_Status
Min.   :2024-02-06      Length:500
1st Qu.:2024-06-09      Class :character 1st Qu.:2024-06-09
Median :2024-10-12      Mode  :character Median :2024-10-12
Mean   :2024-10-12      Mean   :2024-10-12
3rd Qu.:2025-02-14      3rd Qu.:2025-02-14
Max.   :2025-06-10      Max.   :2025-06-10
```

The Environment pane shows the Global Environment with variables: a, bad_rows, cn, df, IQR, missing_rows, missing_rows2, and n.

5. Select two numeric columns(Bill_Amount, Age)

The screenshot shows the RStudio interface. The script editor contains the following code:

```
head(data,15) ----#Task 1.
dim(data) -----#Task 2.
str(data) -----#Task 3.
summary(data)-----#Task 4.
# Randomly insert 100 NA values in Age
data$Age[sample(1:n, 50)]<- NA
data$Age
# Randomly insert 100 NA values in Bill_Amount
data$Bill_Amount[sample(1:n, 50)] <- NA
data$Bill_Amount
data_subset <- data[, c("Age", "Bill_Amount")]
head(data_subset)
view(data)
```

The console output shows the summary of the data frame and the first few rows of the subset:

```
> summary(data)
  Patient_ID Patient_name      Disease      Age      Bill_Amount
Min.   :1000      Length:500      Min.   :18.00      Min.   : 1548
1st Qu.:1000      Class :character 1st Qu.:27.00      1st Qu.: 6348
Median :1000      Mode  :character  Median :37.00      Median :10618
Mean   :1000      Mean   :36.55      Mean   :10693
3rd Qu.:1000      3rd Qu.:46.00      3rd Qu.:15074
Max.   :1000      Max.   :55.00      Max.   :19991

Admission_date Doctor_Assigned Payment_Status
Min.   :2024-02-06      Length:500
1st Qu.:2024-06-09      Class :character 1st Qu.:2024-06-09
Median :2024-10-12      Mode  :character Median :2024-10-12
Mean   :2024-10-12      Mean   :2024-10-12
3rd Qu.:2025-02-14      3rd Qu.:2025-02-14
Max.   :2025-06-10      Max.   :2025-06-10

> head(data_subset)
  Age Bill_Amount
1 37      9288
2 19     17737
3 27     13946
4 24     3201
5 55     10519
6 NA      14618
```

The Environment pane shows the Global Environment with variables: data_subset, df2, dup, hers.dat, mtcars, mydat, mydat.dat, and raw.

6. Insert 100 missing values across these columns.

The screenshot shows the RStudio interface with the following code in the script editor:

```
16 head(data, 15) -----#Task 1.
17
18 dim(data) -----#Task 2.
19
20 str(data) -----#Task 3.
21
22 summary(data)-----#Task 4.
23
24 # Randomly insert 100 NA values in Age
25 data$Age[sample(1:n, 50)]<- NA
26 data$Age
27 # Randomly insert 100 NA values in Bill_Amount
28 data$Bill_Amount[sample(1:n, 50)] <- NA
29
30
31 View(data)
32
33 arrange(data, desc("Bill_Amount"))
34
35
```

The console output shows the first 30 rows of the 'data' object, with the 'Age' column containing many NA values:

```
> data$Age
[1] 37 19 27 24 55 NA 40 NA 41 21 46 NA 28 50 20 50 NA NA 19 37 31 42 24 52 48 NA 33 39 NA
[30] 50 NA NA 41 NA 40 NA 49 NA 43 40 43 NA 25 32 53 20 50 55 43 NA NA 43 NA 40 55 27 55 25 18
[59] 33 19 31 NA 43 25 NA 26 50 20 19 NA 26 NA 30 46 NA 31 33 24 53 NA 42 44 30 NA 27 47 29
[88] 25 24 32 36 40 52 23 24 21 27 34 25 NA NA 31 NA 23 NA 50 NA 54 51 NA NA NA 21 NA 28 NA
[117] 53 NA 52 50 NA 40 NA 38 22 NA NA NA 21 NA NA 18 28 54 28 32 NA 32 NA 42 53 NA 24 21 54
[146] 41 55 21 20 NA 27 46 40 22 41 49 42 28 NA 32 NA 27 NA 52 27 25 49 24 NA 34 NA 50 NA 27
[175] 27 19 32 NA 52 NA NA 26 40 NA 32 39 25 46 NA 26 51 28 NA 24 28 50 28 55 38 27 20 NA 51
[204] 45 30 25 19 23 40 NA 49 31 NA NA NA NA 55 28 34 24 36 NA NA 43 44 34 50 NA NA 44 27 26 48
[233] 36 NA NA 38 46 53 37 40 22 NA 35 29 NA 55 35 27 26 52 NA NA NA NA NA 19 NA 38 48 NA 44
[262] 40 20 32 24 NA 48 35 35 52 NA 32 34 24 47 25 18 NA NA 42 19 28 21 37 NA NA 20 45 20
[291] 45 46 40 32 40 41 50 54 46 23 NA NA NA 41 49 25 33 49 18 47 42 20 NA 34 37 52 NA 54 41 NA
[320] 25 37 NA 39 34 NA NA 38 33 43 52 52 43 30 53 42 45 NA 30 NA 42 46 37 46 30 30 52 22
[349] 30 40 45 25 55 41 33 28 NA 47 49 NA 32 NA 27 39 NA 42 NA NA 44 NA 36 NA 51 40 29 NA 52
[378] 46 NA 46 51 26 18 27 40 NA 49 NA 43 NA NA NA 31 47 NA 38 40 47 42 NA 21 20 NA 45 29 20
```

The screenshot shows the RStudio interface with the following code in the script editor:

```
16 head(data, 15) -----#Task 1.
17
18 dim(data) -----#Task 2.
19
20 str(data) -----#Task 3.
21
22 summary(data)-----#Task 4.
23
24 # Randomly insert 100 NA values in Age
25 data$Age[sample(1:n, 50)]<- NA
26 data$Age
27 # Randomly insert 100 NA values in Bill_Amount
28 data$Bill_Amount[sample(1:n, 50)] <- NA
29 data$Bill_Amount
30
31 View(data)
32
33 arrange(data, desc("Bill_Amount"))
34
35
```

The console output shows the first 30 rows of the 'data\$Bill_Amount' vector, with the first 100 values being NA:

```
> data$Bill_Amount[sample(1:n, 50)] <- NA
> data$Bill_Amount
[1] 9288 17737 13946 3201 10519 14618 5797 16772 16265 18647 NA 4666 17673 6883
[15] NA 2255 18912 14656 11070 16613 14311 5211 7051 7847 16099 2862 14045 14956
[29] 2386 16652 16654 2413 4469 NA 15904 14264 17884 16697 4623 19426 4965 13055
[43] 19675 4963 8716 7298 13465 15977 1570 12442 9341 7641 19626 3288 3145 17627
[57] 3441 18318 11550 6982 13327 8346 9856 10057 11404 7585 6460 7489 13187 NA
[71] 10621 9680 18229 4503 9889 8632 5254 16599 8633 16516 1897 3064 5350 18839
[85] 6855 13747 8647 1768 NA 13191 NA 8934 8678 19263 8033 19650 16063 3595
[99] NA 9434 NA 3936 19531 10810 8823 5354 4119 17574 15032 11756 2968 10244
[113] 9204 16167 4057 3814 3178 7844 17365 6958 18534 11506 13804 15461 7075 15668
[127] 9613 NA 9063 19396 5675 6238 5321 2329 8695 5212 6896 19845 14163 NA
[141] 14420 3559 8135 9843 5242 NA 10397 1883 19991 9549 13514 2690 12347 14990
[155] 13053 10616 NA 8127 19375 NA 18206 11572 12048 19331 15456 8928 10158 6087
[169] NA 17336 3803 7741 10685 7804 2173 11675 15540 12632 9217 4776 6910 2710
```


7. Replace missing values using the mean column.

The screenshot shows the RStudio interface. The script editor contains the following code:

```
summary(data)
# Randomly insert 100 NA values in Age
data$Age[sample(1:n, 50)] <- NA
data$Age
# Randomly insert 100 NA values in Bill_Amount
data$Bill_Amount[sample(1:n, 50)] <- NA
data$Bill_Amount
data_subset <- data[, c("Age", "Bill_Amount")]
head(data_subset)
data$Age[is.na(data$Age)] <- mean(data$Age, na.rm = TRUE)
data$Age
data$Bill_Amount[is.na(data$Bill_Amount)] <- mean(data$Bill_Amount, na.rm = TRUE)
data$Bill_Amount
```

The console shows the output of the summary function, displaying the distribution of values for Age and Bill_Amount after the replacement.

8. Rename columns:

- Patient_name -> Client_Name
- Admission_Date -> Visit_Date
- Payment_Status -> Billing_Status

The screenshot shows the RStudio interface. The script editor contains the following code:

```
data$Bill_Amount[is.na(data$Bill_Amount)] <- mean(data$Bill_Amount, na.rm = TRUE)
data$Bill_Amount
colnames(data)[colnames(data) == "Patient_Name"] <- "Client_Name"
colnames(data)[colnames(data) == "Admission_Date"] <- "Visit_Date"
colnames(data)[colnames(data) == "Payment_Status"] <- "Billing_Status"
head(data)
```

The console shows the output of the head function, displaying the first few rows of the dataset with the updated column names.

9. Move the last column to the first positions

The screenshot shows the RStudio interface. In the script editor, the following code is executed:

```
data$Age  
data$Bill_Amount[is.na(data$Bill_Amount)] <- mean(data$Bill_Amount, na.rm = TRUE)  
data$Bill_Amount  
colnames(data)[colnames(data) == "Patient_Name"] <- "Client_Name"  
colnames(data)[colnames(data) == "Admission_Date"] <- "Visit_Date"  
colnames(data)[colnames(data) == "Payment_Status"] <- "Billing_Status"  
head(data)  
data <- data[, c(ncol(data), 1:(ncol(data)-1))]  
head(data)
```

The console output shows the result of the code:

```
R - R 4.5.2 ~ ~/Rpractic02/  
> data <- data[, c(ncol(data), 1:(ncol(data)-1))]  
> head(data)  
Billing_Status Patient_ID Patient_name Disease Age Bill_Amount Admission_date  
1 Not Completed 1000 Patient 1 Typhoid 37.00000 9288 2024-02-06  
2 Completed 1000 Patient 2 Typhoid 19.00000 17737 2024-02-07  
3 Not Completed 1000 Patient 3 Typhoid 27.00000 13946 2024-02-08  
4 Half Due 1000 Patient 4 Heart attack 24.00000 3201 2024-02-09  
5 Completed 1000 Patient 5 Typhoid 55.00000 10519 2024-02-10  
6 Not Completed 1000 Patient 6 Heart attack 36.39617 14618 2024-02-11  
Doctor_Assigned  
1 Hemant  
2 Sumit  
3 Aasha  
4 Sumit  
5 Sumit
```

The Environment pane on the right shows the following objects:

Object	Size
data	500 obs. of 8 variables
data_subset	500 obs. of 2 variables
df2	11 obs. of 2 variables
dup	15 obs. of 7 variables
hers.dat	2763 obs. of 40 variables
mtcars	32 obs. of 11 variables
mydat	10 obs. of 5 variables
mydat.dat	2763 obs. of 40 variables

Subsetting/ Filtering Tasks

Perform all of the following

1. Patients with Bill_Amount>5000

The screenshot shows the RStudio interface. In the script editor, the following code is executed:

```
filter(data, Bill_Amount > 5000)  
filter(data, Age > 50)  
filter(data, Billing_Status == "Paid")  
data$Paid <- ifelse(data$Billing_Status == "Paid", 1, 0)  
filter(data, Age > 40 & Bill_Amount > 10000)  
filter(data, Disease == "Diabetes" | Disease == "Heart")
```

The console output shows the result of the code:

```
R - R 4.5.2 ~ ~/Rpractic02/  
> filter(data, Bill_Amount > 5000)  
Billing_Status Patient_ID Patient_name Disease Age Bill_Amount Admission_date  
1 Not Completed 1000 Patient 1 Typhoid 37.00000 9288.00 2024-02-06  
2 Completed 1000 Patient 2 Typhoid 19.00000 17737.00 2024-02-07  
3 Not Completed 1000 Patient 3 Typhoid 27.00000 13946.00 2024-02-08  
4 Completed 1000 Patient 5 Typhoid 55.00000 10519.00 2024-02-10  
5 Not Completed 1000 Patient 6 Heart attack 36.39617 14618.00 2024-02-11  
6 Half Due 1000 Patient 7 Heart attack 40.00000 5797.00 2024-02-12  
7 Completed 1000 Patient 8 Heart attack 36.39617 16772.00 2024-02-13  
8 Half Due 1000 Patient 9 Typhoid 41.00000 16265.00 2024-02-14  
9 Half Due 1000 Patient 10 Diabetes 21.00000 18647.00 2024-02-15  
10 Half Due 1000 Patient 11 Heart attack 46.00000 10725.76 2024-02-16  
11 Completed 1000 Patient 13 Diabetes 28.00000 17673.00 2024-02-18  
12 Half Due 1000 Patient 14 Heart attack 50.00000 6883.00 2024-02-19  
13 Not Completed 1000 Patient 15 Typhoid 20.00000 10725.76 2024-02-20
```

The Environment pane on the right shows the following objects:

Object	Size
data	500 obs. of 8 variables
data_subset	500 obs. of 2 variables
df2	11 obs. of 2 variables
dup	15 obs. of 7 variables
hers.dat	2763 obs. of 40 variables
mtcars	32 obs. of 11 variables
mydat	10 obs. of 5 variables
mydat.dat	2763 obs. of 40 variables

2. Patients with Age>50


```

46 filter(data, Bill_Amount > 5000)
47
48 filter(data, Age > 50)
49
50 filter(data, Billing_Status == "Paid")
51
52 data$Paid <- ifelse(data$Billing_Status == "Paid", 1, 0)
53
54 filter(data, Age > 40 & Bill_Amount > 10000)
55
56 filter(data, Disease == "Diabetes" | Disease == "Heart")
57
58
59
60
61
62
63
64
65
66 (Top Level)

```

Billing_Status	Patient_ID	Patient_name	Disease	Age	Bill_Amount	Admission_date
Completed	1000	Patient 5	Typhoid	55	10519.00	2024-02-10
Half Due	1000	Patient 24	Heart attack	52	7847.00	2024-02-29
Not Completed	1000	Patient 45	Typhoid	53	8716.00	2024-03-21
Half Due	1000	Patient 48	Heart attack	55	15977.00	2024-03-24
Half Due	1000	Patient 54	Diabetes	55	3288.00	2024-03-30
Completed	1000	Patient 56	Diabetes	55	17627.00	2024-04-01
Half Due	1000	Patient 79	Diabetes	53	8633.00	2024-04-24
Completed	1000	Patient 93	Diabetes	52	8678.00	2024-05-08
Completed	1000	Patient 108	Heart attack	54	17574.00	2024-05-23
Not Completed	1000	Patient 109	Heart attack	51	15032.00	2024-05-24
Half Due	1000	Patient 117	Heart attack	53	3178.00	2024-06-01
Not Completed	1000	Patient 119	Heart attack	52	17365.00	2024-06-03
Completed	1000	Patient 134	Heart attack	54	2329.00	2024-06-18

3. Patients with Payment_status = "Paid"

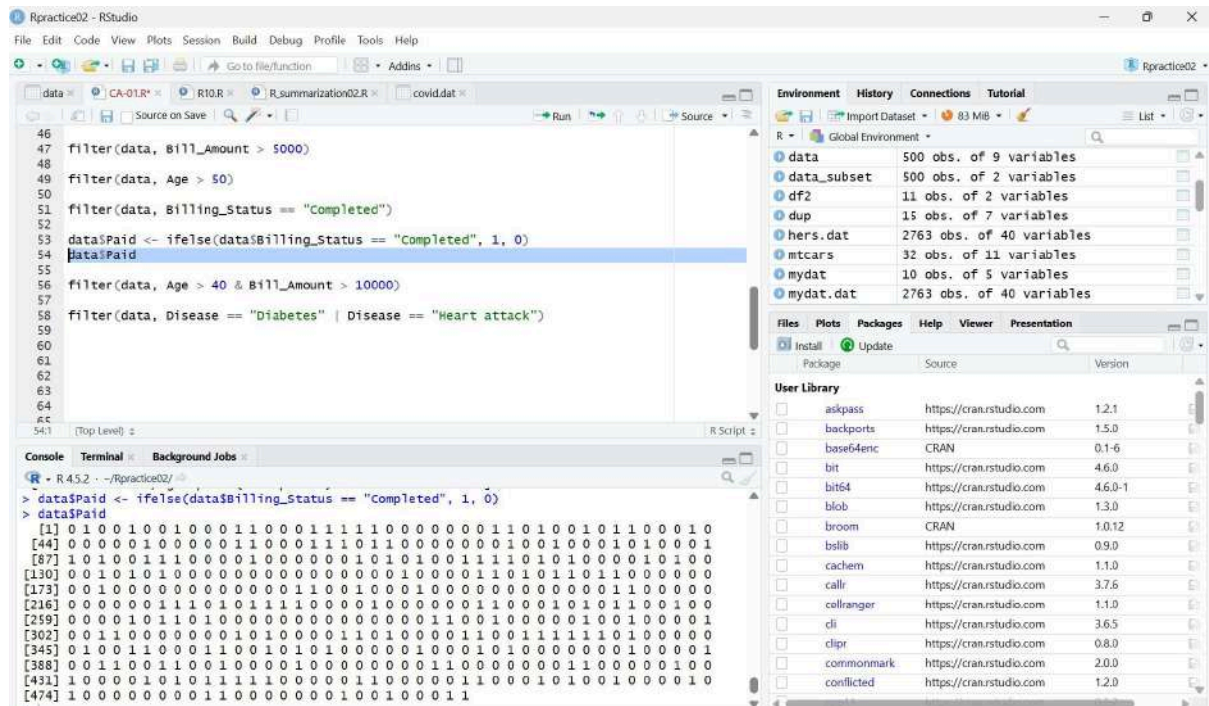
```

46 filter(data, Bill_Amount > 5000)
47
48 filter(data, Age > 50)
49
50 filter(data, Billing_Status == "Completed")
51
52 data$Paid <- ifelse(data$Billing_Status == "Completed", 1, 0)
53
54 filter(data, Age > 40 & Bill_Amount > 10000)
55
56 filter(data, Disease == "Diabetes" | Disease == "Heart attack")
57
58
59
60
61
62
63
64
65
66 (Top Level)

```

Billing_Status	Patient_ID	Patient_name	Disease	Age	Bill_Amount	Admission_date
Completed	1000	Patient 2	Typhoid	19.00000	17737.00	2024-02-07
Completed	1000	Patient 5	Typhoid	55.00000	10519.00	2024-02-10
Completed	1000	Patient 8	Heart attack	36.39617	16772.00	2024-02-13
Completed	1000	Patient 12	Heart attack	36.39617	4666.00	2024-02-17
Completed	1000	Patient 13	Diabetes	28.00000	17673.00	2024-02-22
Completed	1000	Patient 17	Typhoid	36.39617	18912.00	2024-02-22
Completed	1000	Patient 18	Typhoid	36.39617	14656.00	2024-02-23
Completed	1000	Patient 19	Diabetes	19.00000	11070.00	2024-02-24
Completed	1000	Patient 20	Diabetes	37.00000	16613.00	2024-02-25
Completed	1000	Patient 21	Diabetes	31.00000	14311.00	2024-02-26
Completed	1000	Patient 29	Typhoid	36.39617	2386.00	2024-03-05
Completed	1000	Patient 30	Heart attack	50.00000	16652.00	2024-03-06

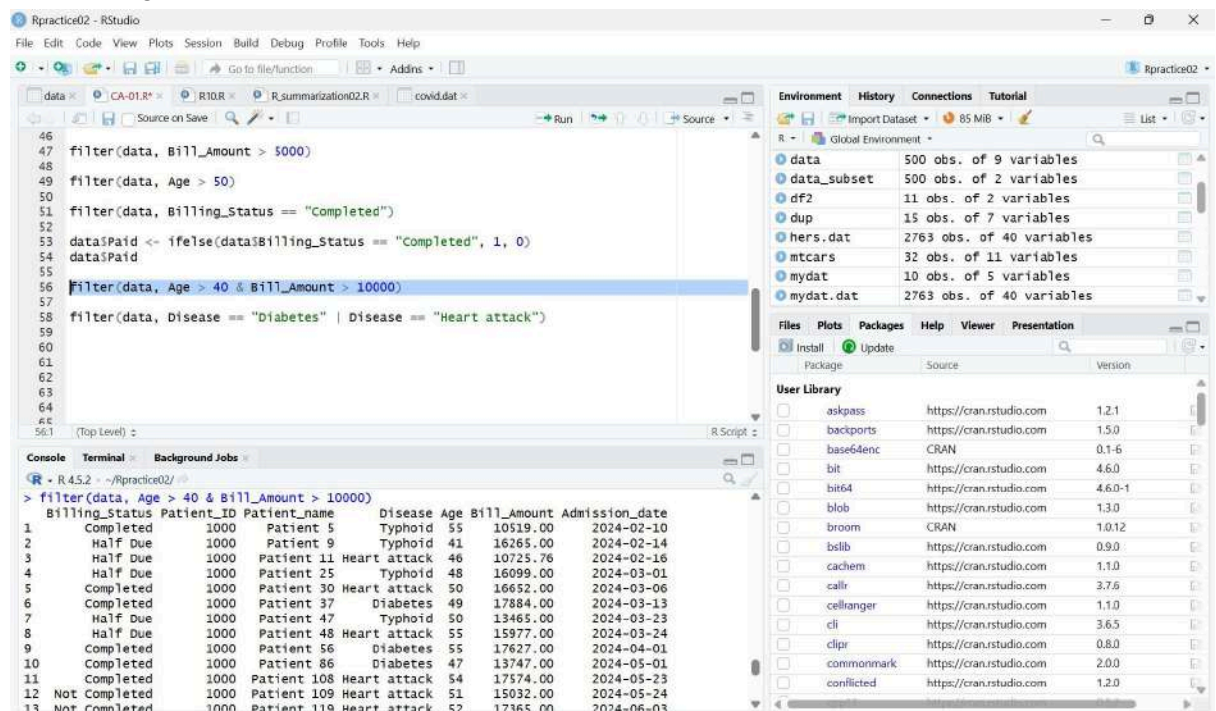
4. Add new Column paid



5. Multiple Condition filters

6. Patients where:

- Age>40 AND Bill_Amount>10000



7. Patients where:

- Disease="Diabetes" OR Disease="Heart"

Rpractice02 - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

data CA-01.R R10.R R_summarization02.R covid.dat

Run Source

```
46 filter(data, Bill_Amount > 5000)
47
48 filter(data, Age > 50)
49
50 filter(data, Billing_Status == "Completed")
51
52 data$Paid <- ifelse(data$Billing_Status == "Completed", 1, 0)
53 data$Paid
54
55 filter(data, Age > 40 & Bill_Amount > 10000)
56
57 filter(data, Disease == "Diabetes" | Disease == "Heart attack")
58
59
60
61
62
63
64
65
66 (Top Level)
```

Environment History Connections Tutorial

Global Environment

data	500 obs. of 9 variables
data_subset	500 obs. of 2 variables
df2	11 obs. of 2 variables
dup	15 obs. of 7 variables
hers.dat	2763 obs. of 40 variables
mtcars	32 obs. of 11 variables
mydat	10 obs. of 5 variables
mydat.dat	2763 obs. of 40 variables

Files Plots Packages Help Viewer Presentation

Install Update

Package Source Version

User Library

askpass	https://cran.rstudio.com	1.2.1
backports	https://cran.rstudio.com	1.5.0
base64enc	CRAN	0.1-6
bit	https://cran.rstudio.com	4.6.0
bit64	https://cran.rstudio.com	4.6.0-1
blob	https://cran.rstudio.com	1.3.0
broom	CRAN	1.0.12
bslib	https://cran.rstudio.com	0.9.0
cachem	https://cran.rstudio.com	1.1.0
callr	https://cran.rstudio.com	3.7.6
cellranger	https://cran.rstudio.com	1.1.0
cli	https://cran.rstudio.com	3.6.5
clipr	https://cran.rstudio.com	0.8.0
commonmark	https://cran.rstudio.com	2.0.0
conflicted	https://cran.rstudio.com	1.2.0

Console Terminal Background Jobs

R - R 4.5.2 - ~/Rpractice02/

```
> filter(data, Disease == "Diabetes" | Disease == "Heart attack")
  Billing_Status Patient_ID Patient_name Disease Age Bill_Amount Admission_date
1 Half Due 1000 Patient 4 Heart attack 24.00000 3201.00 2024-02-09
2 Not Completed 1000 Patient 6 Heart attack 36.39617 14618.00 2024-02-11
3 Half Due 1000 Patient 7 Heart attack 40.00000 5797.00 2024-02-12
4 Completed 1000 Patient 8 Heart attack 36.39617 16772.00 2024-02-13
5 Half Due 1000 Patient 10 Diabetes 21.00000 18647.00 2024-02-15
6 Half Due 1000 Patient 11 Heart attack 46.00000 10725.76 2024-02-16
7 Completed 1000 Patient 12 Heart attack 36.39617 4666.00 2024-02-17
8 Completed 1000 Patient 13 Diabetes 28.00000 17673.00 2024-02-18
9 Half Due 1000 Patient 14 Heart attack 50.00000 6883.00 2024-02-19
10 Half Due 1000 Patient 16 Diabetes 50.00000 2255.00 2024-02-21
11 Completed 1000 Patient 19 Diabetes 19.00000 11070.00 2024-02-24
12 Completed 1000 Patient 20 Diabetes 37.00000 16613.00 2024-02-25
```

